

Jiaqi Zhang

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EDUCATION

McGill University | Montreal, QC

Expected May 2028

B.Sc. in Mathematics and Computer Science

Relevant Coursework: Linear Algebra, Probability, Abstract Algebra, Algorithms, Real Analysis

RESEARCH INTERESTS

Cryptographic protocols; threshold cryptography; secure computation; rational and adversarial behavior

RESEARCH

Verify-or-Scar: Bribery Resistance for Weighted Threshold Services

Manuscript under review

Independently completed the research and manuscript, with guidance from Prof. Honghao Fu.

- Studied bribery in weighted threshold services where an attacker can verify a disclosed contribution before deciding whether to purchase it.
- Designed a mechanism in which trusted external verification triggers persistent liability, including when a verified contribution is not purchased.
- Analyzed procurement and participation incentives under rational behavior, including conditions that make procurement more costly or infeasible.
- Implemented and evaluated the mechanism in a trusted-execution setting, testing sanction persistence, retries, recovery, non-purchase, and threshold composition.

Adrian Vetta Research Group, McGill University

Research Group Participant

- Participate in research discussions on algorithmic game theory, incentives, and strategic behavior.

SELECTED PROJECTS

Road Sign Classification with Denoising Pipeline

Supervised by Prof. David Woodruff, Carnegie Mellon University

- Developed a noise-aware classification pipeline with multi-stage denoising and feature enhancement to study the effect of preprocessing on model performance.
- Analyzed noise characteristics and misclassification patterns to guide preprocessing and algorithmic improvements.

Formal Modeling of Individual and Collective Utility Tradeoffs

- Used sets, mappings, and logical formalization to model tradeoffs between individual incentives and collective outcomes.
- Examined when individual sacrifice leads to collective benefit and analyzed incentive structures for logical consistency and edge cases.

TECHNICAL SKILLS

Programming: Python, Java, NumPy

Security & Systems: Cryptographic protocol analysis, threshold systems, trusted execution environments (TEEs), remote attestation, Azure TDX

Research & Experimentation: Mathematical and security modeling, algorithmic analysis, protocol implementation, experimental design, failure and recovery testing